

Performance Testing

Date	3 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask Test Server
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Loan Eligibility Prediction API (/predict)
Test Environment	Local System (Python 3.x, Flask)
Test Date	4 july 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single loan prediction request	1	30	Prediction generated successfully
2	Multiple prediction requests	10	60	Stable performance without errors
3	Invalid user input	5	30	Validation error displayed
4	Continuous prediction requests	20	120	Application remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.2 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 sec	2.8 sec	Pass	Within target
3	Throughput (Req/sec)	10 Req/sec	15 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	45%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	52%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings :

The Smart Lender application successfully generated loan approval predictions with low response time and consistent performance. The XGBoost model accurately classified eligible and non-eligible applicants based on the provided input features.

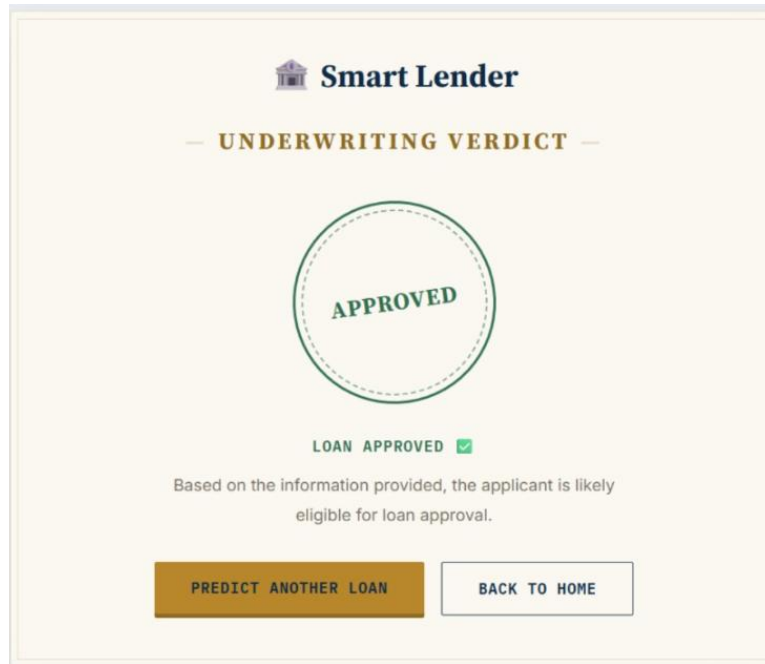
Bottlenecks Identified:

No major performance bottlenecks were observed during testing. The application remained stable under normal usage conditions.

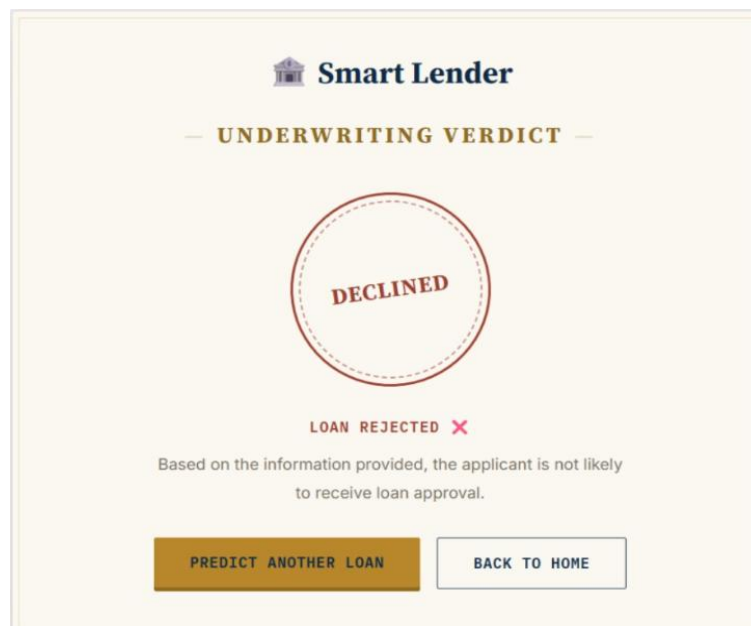
Optimization Steps Taken:

The trained XGBoost model was loaded once during application startup, user inputs were validated before prediction, and preprocessing was optimized to minimize response time.

Step 5: Screenshots / Evidence



Screenshot 1: Loan Approved Prediction



Screenshot 2: Loan Rejected Prediction